

Fabrication and characterization of electron microscope 3D resolution standards

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Abstract

Scanning Electron Microscopy (SEM) plays a critical role in nanoscale imaging and analysis, supporting applications in materials science, microelectronics, and nanofabrication. A fundamental parameter that determines the utility of SEM is its spatial resolution, which is the ability to distinguish fine structural details. To evaluate and calibrate this performance effectively, resolution standard samples with well-defined and reproducible features are essential.

This project focuses on the development of a more precise grid pattern resolution standard sample for SEM resolution evaluation. The design takes into steep-edged metallic structures, which serve as reliable references for measuring image sharpness and beam focus. To achieve this goal, lithography technology is used to define fine patterns, followed by controlled metal deposition. Special attention should be paid to minimizing edge related artifacts that may affect resolution measurements.

By analyzing images of the fabricated structures, the resolution of the microscope can be quantitatively assessed through intensity profiles and edge response metrics such as full width at half maximum. The resulting standard sample demonstrates high reproducibility, clear edge contrast, and strong compatibility with high-resolution SEM systems.

Overall, the project has produced a precision structured SEM resolution standard sample, contributing to the evaluation and calibration of electron microscope performance and standardized data.

Table of Contents

1 Introduction	3
1.1 Background Introduction	3
1.2 Project Plan	4
1.3 Design Significance	5
1.4 Experimental Scheme	6
2 Lithography	6
2.1 Basic principles and Methods of Lithography	6
2.2 Clean Room Environment in Resist Processing	7
2.3 Sputtering metal layer	8
2.4 Spin coating	11
2.5 Softbake	22
2.6 Exposure	23
2.7 Development	28
2.8 Metal Plating	30
2.9 Lift-off	38
3 Image Analysis	41
3.1 Thickness measurement	41
3.2 Sturcture measurement	44
3.3 Weakness analysis of samples	49
4 Data Analysis	51
4.1 Analysis method	51
4.2 Proton beam measurement of sample 4	54
4.3 Error analysis	59
4.4 SEM measurement of sample 4	60
4.5 Sample as a Beam Focus Reference	65
5 Conclusion	66
6 References	67